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Studierichting: Informatica
Oefeningenreeks 2D nr. 12.9

$f(x) = \sqrt{2x^2+3} - x$

$\lim_{x\to\infty} (\sqrt{2x^2+3} - x)$

$\lim_{x\to-\infty} (\sqrt{2x^2+3} - x) = +\infty$

$\lim_{x\to+\infty} (\sqrt{2x^2+3} - x) = (+\infty) - (+\infty)$

$= \lim_{x\to+\infty} \frac{2x^2+3-x^2}{\sqrt{2x^2+3}+x}$

$= \lim_{x\to+\infty} \frac{x^2+3}{\sqrt{2x^2+3}+x}$

$= \lim_{x\to+\infty} \frac{x\left(x+\frac{3}{x}\right)}{x\left(\sqrt{2+\frac{3}{x^2}}+1\right)}$

$= \lim_{x\to+\infty} \frac{+\infty}{\sqrt{2}+1} = +\infty$

Dus $\lim_{x\to\infty} \sqrt{2x^2+3} - x = +\infty$

References